SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Gravitation – Level 3 (Advanced)

PRACTICE SHEET – 3

Gravitation – Level 3 (Advanced): Practice Sheet 3 of 3

Name: _____	Class / Section: _____	Date: _____
Time Allowed: 60 Mins	**Total Questions: 15**	Roll No.: _____

Instructions: Attempt all 15 questions. Questions with options have only one correct option; the others need full working. Sections: A – Acceleration due to Gravity, B – Gravitational Potential and Potential Energy, C – Motion of Satellite, D – Newton's Law, Gravitational Field and Kepler's Laws. Time Allowed: 60 Mins.

Section A – Acceleration due to Gravity

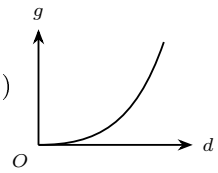
Q.1 The acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the pole is ω. An object is weighed at the equator and at a height h above the poles by using a spring balance. If the weights are found to be same, then h is: ($h \ll R$, where R is the radius of the earth)

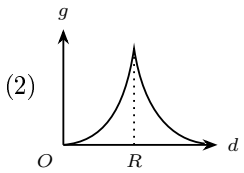
(1) $\dfrac{R^2\omega^2}{g}$ (2) $\dfrac{R^2\omega^2}{4g}$

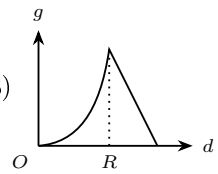
(3) $\dfrac{R^2\omega^2}{2g}$ (4) $\dfrac{R^2\omega^2}{8g}$

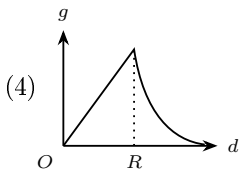
[Nucleus, JM Q.28 [JEE Main-2020]]

Q.2 The variation of acceleration due to gravity g with distance d from centre of the earth is best represented by (R = Earth's radius):

(1) 

(2) 

(3) 

(4) 

[Nucleus, JM Q.10 [JEE Main 2017]]

Q.3 Calculate the distance from the surface of the earth at which above and below the surface acceleration due to gravity is the same.

[Motion, Ex-III Q.8]

Q.4 The Earth may be regarded as a spherically shaped uniform core of density ρ_1 and radius $R/2$ surrounded by a uniform shell of thickness $R/2$ and density ρ_2. Find the ratio $\dfrac{\rho_1}{\rho_2}$ if the value of acceleration due to gravity is the same at surface as at depth $R/2$ from the surface.

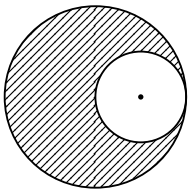
[Nucleus, S-1 Q.6]

Section B – Gravitational Potential and Potential Energy

Q.5 An object is projected vertically upward from the surface of the earth of mass M with a velocity such that the maximum height reached is eight times the radius R of the earth. Calculate:
(i) the initial speed of projection
(ii) the speed at half the maximum height.

[Motion, Ex-III Q.3]

Q.6 From a solid sphere of mass M and radius R, a spherical portion of radius $R/2$ is removed, as shown in the figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the centre of the cavity thus formed is (G = gravitational constant)



(1) $-\dfrac{GM}{R}$ (2) $-\dfrac{2GM}{3R}$

(3) $-\dfrac{2GM}{R}$ (4) $-\dfrac{GM}{2R}$

[Nucleus, JM Q.8 [JEE Main 2015]]

Q.7 A particle is fired vertically from the surface of the earth with a velocity kv_e, where v_e is the escape velocity and $k < 1$. Neglecting air resistance and assuming earth's radius as R_e. Calculate the height to which it will rise from the surface of the earth.

[Motion, Ex-III Q.10]

Q.8 Find the potential energy of a system of eight particles placed at the vertices of a cube of side L. Neglect the self energy of the particles.

[Motion, Ex-III Q.12]

Section C – Motion of Satellite

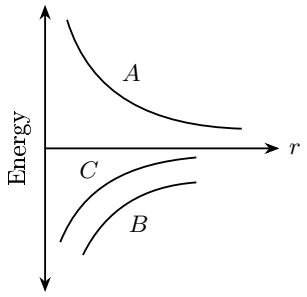
Q.9 A satellite of 10^3 kg mass is revolving in circular orbit of radius $2R$. If $\dfrac{10^4 R}{6}$ J energy is supplied to the satellite, it would revolve in a new circular orbit of radius (use $g = 10$ m/s^2, R = radius of earth)

(A) 4R (B) 6R

(C) 2.5R (D) 3R

[JEE Main 2024, 9th April Evening Shift]

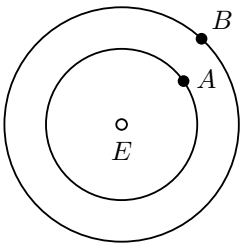
Q.10 The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct statement about the curves A, B and C



(A) A shows the kinetic energy, B the total energy and C the potential energy of the system
(B) C shows the total energy, B the kinetic energy and A the potential energy of the system
(C) C and A are kinetic and potential energies respectively and B is the total energy of the system
(D) A and B are kinetic and potential energies and C is the total energy of the system

[Motion, Ex-I Q.22]

Q.11 Two satellites A and B of masses 200 kg and 400 kg are revolving round the earth at height of 600 km and 1600 km respectively. If T_A and T_B are the time periods of A and B respectively then the value of $T_B - T_A$:

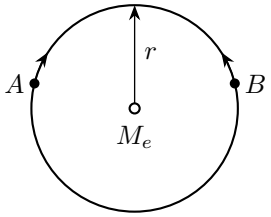


[Given: radius of earth = 6400 km, mass of earth $= 6 \times 10^{24}$ kg]

(1) 1.33×10^3 s (2) 3.33×10^2 s
(3) 4.24×10^3 s (4) 4.24×10^2 s

[Nucleus, JM Q.35 [JEE Main-2021]]

Q.12 Consider two satellites A and B of equal mass m, moving in the same circular orbit of radius r around the earth E but in opposite sense of rotation and therefore on a collision course (see figure).

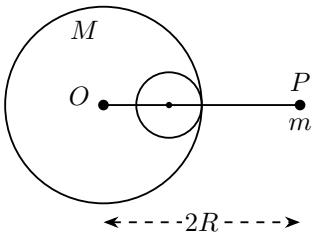


(a) In terms of G, M_e, m and r find the total mechanical energy $E_A + E_B$ of the two satellite plus earth system before collision.
(b) If the collision is completely inelastic so that wreckage remains as one piece of tangled material (mass = $2m$), find the total mechanical energy immediately after collision.
(c) Describe the subsequent motion of the wreckage.

[Motion, Ex-III Q.9]

Section D – Newton's Law, Gravitational Field and Kepler's Laws

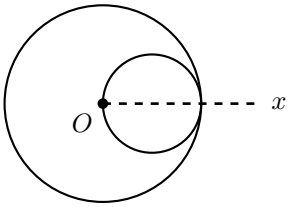
Q.13 A small point mass m is placed at a distance $2R$ from the centre 'O' of a big uniform solid sphere of mass M and radius R. The gravitational force on 'm' due to M is F_1. A spherical part of radius $R/3$ is removed from the big sphere as shown in the figure and the gravitational force on m due to remaining part of M is found to be F_2. The value of ratio $F_1 : F_2$ is



(A) 11 : 10 (B) 12 : 11
(C) 16 : 9 (D) 12 : 9

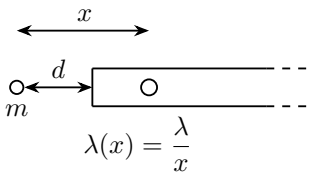
[JEE Main 2025, 22nd January Morning Shift]

Q.14 A sphere of radius R has its centre at the origin. It has a uniform mass density ρ_0 except that there is a spherical hole of radius $r = R/2$ whose centre is at $x = R/2$ as in fig.
(a) Find gravitational field at points on the axis for $x > R$.
(b) Show that the gravitational field inside the hole is uniform, find its magnitude and direction.



[Motion, Ex-III Q.15]

Q.15 Find the gravitational force of interaction between the mass m and an infinite rod of varying mass density λ such that $\lambda(x) = \lambda/x$, where x is the distance from mass m. Given that mass m is placed at a distance d from the end of the rod on its axis as shown in figure.



[Motion, Ex-IV Q.3]

ANSWER KEY – Practice Sheet 3

Answers are taken from the source answer keys: Motion (Answer Ex-I, Ex-III, Ex-IV), Nucleus (Answer Key page; JM answers are option numbers) and the JEE PYQ file (Answer Key at the end).

1. [Nucleus JM Q.28] (3)

2. [Nucleus JM Q.10] (4)

3. [Motion Ex-III Q.8] $h = \dfrac{\sqrt{5}-1}{2}\,R$

4. [Nucleus S-1 Q.6] $\dfrac{7}{3}$

5. [Motion Ex-III Q.3] (i) $\dfrac{4}{3}\sqrt{\dfrac{GM}{R}}$; (ii) $\dfrac{2}{3}\sqrt{\dfrac{2GM}{5R}}$

6. [Nucleus JM Q.8] (1)

7. [Motion Ex-III Q.10] $\dfrac{R_e\,k^2}{1-k^2}$

8. [Motion Ex-III Q.12] $-\dfrac{4GM^2}{L}\left[3+\dfrac{3}{\sqrt{2}}+\dfrac{1}{\sqrt{3}}\right]$

9. [JEE PYQ file Q.18] B

10. [Motion Ex-I Q.22] D

11. [Nucleus JM Q.35] (1)

12. [Motion Ex-III Q.9] (a) $-\dfrac{GmM_e}{r}$; (b) $-\dfrac{2GmM_e}{r}$; (c) not given in the source key

13. [JEE PYQ file Q.17] B

14. [Motion Ex-III Q.15] (a) $\vec{g} = +\dfrac{\pi G\rho_0 R^3}{6}\left[\dfrac{1}{\left(x-\frac{R}{2}\right)^2}-\dfrac{8}{x^2}\right]\hat{i}$; (b) $\vec{g} = -\dfrac{2\pi G\rho_0 R}{3}\,\hat{i}$

15. [Motion Ex-IV Q.3] $\dfrac{Gm\lambda}{2d^2}$

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SELF ASSESSMENT – Practice Sheet 3

Fill this in after checking your answers with the Answer Key.

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Start time: _____	End time: _____
Finished within the allowed time? ☐ Yes ☐ No	(extra / saved time: _____ Mins)

Total	Attempted	Correct	Incorrect	Not attempted
15				

Score (Correct / 15): _____ Percentage: _____ %

Question-wise tracker write ✓ (correct), ✗ (incorrect) or – (not attempted)

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	

Reason for the incorrect / unattempted questions (write question numbers)

Concept not clear: _____	Formula forgotten: _____
Calculation / silly mistake: _____	Ran out of time: _____

Personal feedback guide (tick the band that matches your score)

	Correct	What to do next
☐	14–15	Excellent. Concept and speed are both strong. Move to harder JEE-level problems and keep revising formulae once a week.
☐	11–13	Good. Re-do every incorrect question without looking at the solution, then note the formula or idea you missed.
☐	7–10	Fair. Revise the theory of escape speed, orbital speed, energies of a satellite and Kepler's laws, then re-attempt this sheet after 3 days.
☐	0–6	Needs attention. Re-read the chapter notes and solved examples first, then attempt this sheet again in two parts of 30 mins.

My personal feedback / action plan:

_____

_____

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____